Study Material - Generated by AI

CS3301 - Data Structures

Topic: bubble sort

1. SUMMARY

Bubble sort is a simple sorting algorithm that repeatedly steps through the list, compares adjacent elements and swaps them if they are in the wrong order. The pass through the list is repeated until the list is sorted. The algorithm, which is comparable to a selection sort algorithm, works well on lists that are already partially sorted.

2. SHORT NOTES

SORTING: BUBBLE SORT

Selection sort is not mentioned in the context of a particular sorting algorithm, but rather 'Bubble Sort' is the sorting algorithm that is mentioned in the context of the reference material.

Bubble sort begins at the beginning of the list and compares each pair of adjacent items, if the first item is less than the second item, then the two items are swapped; otherwise no swap is made. When it reaches the end of the list, the process begins again and is repeated until there are no more swaps made; thus the list is sorted.

Bubble sort is an in-place sorting algorithm and it uses the comparison function to sort the list. Bubble sort works well for lists that are already partially sorted. It does n-1 passes of the list where n is the number of items. The last swap gives the smallest last element in order. In each pass n-1 swaps may be performed. Thus the total number of swaps performed in any case will be not more than (n-1) * (n-2) /2 (n-3) /2, etc. Hence the time complexity is of the order of n*n.

TIME COMPLEXITIES

Time complexity refers to the amount of time a particular algorithm takes to complete in terms of the size of the input. If an algorithm has a time complexity of the order of n, this means that it could take anywhere from n to 3 * n (for example) to complete. In the context of bubble sort with a list of n items, the time complexity is of the order of n*n. This means the time complexity for bubble sort is a quadratic time complexity with respect to the size of the list.

BUBBLE SORT EXAMPLE

The steps for bubble sort can be illustrated as follows. We consider an initially unsorted array of integers: A = { 3, 4, 6, 7, 2, 1, 5 }. We begin at the start of the array, comparing each pair of adjacent items and swapping them if necessary. The first pass results in: A = { 3, 4, 6, 7, 2, 1, 5 }, where 3 and 4 are adjacent, 4 and 6 are adjacent, etc. After one pass we get: A = { 3, 2, 4, 6, 7, 1, 5 }. We continue until there are no more swaps made.

3. IMPORTANT QUESTIONS

1. What is the time complexity of bubble sort with respect to the size of the list?

a) n*n

b) n^2

c) 2^n

d) n!

2. What is bubble sort?

a) Simple sorting algorithm

b) Simple file transfer algorithm

c) Simple search algorithm

d) Simple data structure algorithm

3. What is the main difference between bubble sort and selection sort?

a) Selection sort is used for sorting lists of integers, while bubble sort is used for sorting lists of characters.

b) Bubble sort works well for lists that are already partially sorted, while selection sort does not.

c) Bubble sort starts at the beginning of the list, while selection sort starts at the end of the list.

d) Selection sort is an in-place sorting algorithm, while bubble sort is not.

4. What is the purpose of the last step in bubble sort, where the list is repeated until there are no more swaps made?

a) To ensure the list is sorted

b) To make the algorithm more complex

c) To simplify the algorithm

d) To make the list partially sorted

5. What is the term for the algorithm used by bubble sort to compare adjacent items and swap them if necessary?

a) Selection function

b) Comparison function

c) Swap function

d) Sorting function

6. Does bubble sort work well for lists that are already partially sorted?

a) Yes

b) No

c) It depends on the size of the list

d) It depends on the number of swaps made

7. What is the time complexity of bubble sort when the list is already sorted?

a) n

b) n^2

c) 2^n

d) n!

8. What is the time complexity of bubble sort when the list is reversed sorted?

a) n

b) n^2

c) 2^n

d) n!

9. What is the time complexity of bubble sort when the list is partially sorted?

a) n

b) n^2

c) 2^n

d) n!

10. What is the average time complexity of bubble sort?

a) n

b) n^2

c) 2^n

d) n!

4. MCQs

Q1. What is the main characteristic of a linked list?

a) It is a dynamic data structure

b) It is a static data structure

c) It is a collection of items in a sequence

d) It is a collection of items in a non-sequence

Answer: a) It is a dynamic data structure

Q2. What is the term for a data structure that is used to store a collection of items in a sequence?

a) Array

b) Linked list

c) Stack

d) Queue

Answer: b) Linked list

Q3. What is the main advantage of a linked list over an array?

a) It is faster

b) It is slower

c) It is more flexible

d) It is less flexible

Answer: c) It is more flexible

Q4. What is the term for a node in a linked list?

a) Element

b) Item

c) Node

d) Record

Answer: c) Node

Q5. What is the purpose of the link field in a node of a linked list?

a) To point to the next node

b) To point to the previous node

c) To point to the first node

d) To point to the last node

Answer: a) To point to the next node

Q6. What is the time complexity of bubble sort with respect to the size of the list?

a) n

b) n^2

c) 2^n

d) n!

Answer: b) n^2

Q7. What is the main difference between bubble sort and selection sort?

a) Selection sort is used for sorting lists of integers, while bubble sort is used for sorting lists of characters.

b) Bubble sort works well for lists that are already partially sorted, while selection sort does not.

c) Bubble sort starts at the beginning of the list, while selection sort starts at the end of the list.

d) Selection sort is an in-place sorting algorithm, while bubble sort is not.

Answer: b) Bubble sort works well for lists that are already partially sorted, while selection sort does not.

Q8. What is the purpose of the last step in bubble sort, where the list is repeated until there are no more swaps made?

a) To ensure the list is sorted

b) To make the algorithm more complex

c) To simplify the algorithm

d) To make the list partially sorted

Answer: a) To ensure the list is sorted

Q9. What is the term for the algorithm used by bubble sort to compare adjacent items and swap them if necessary?

a) Selection function

b) Comparison function

c) Swap function

d) Sorting function

Answer: b) Comparison function

Q10. Does bubble sort work well for lists that are already partially sorted?

a) Yes

b) No

c) It depends on the size of the list

d) It depends on the number of swaps made

Answer: a) Yes